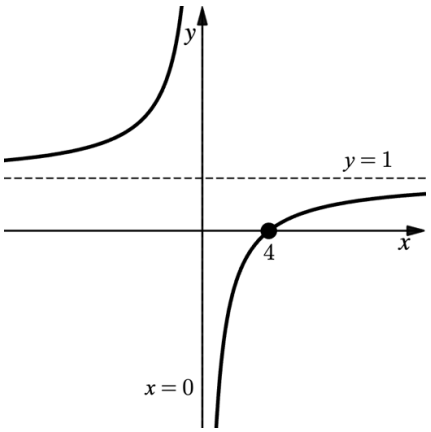
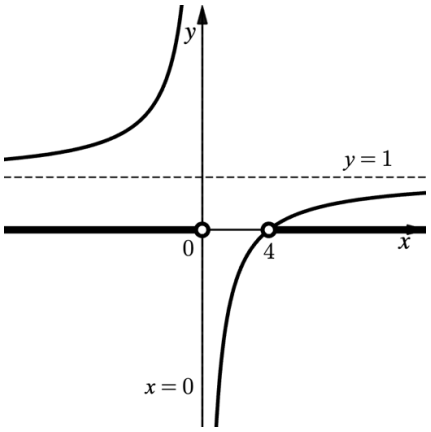
1.17 Inequalities: Rational Functions

Worked Solutions

1 a)

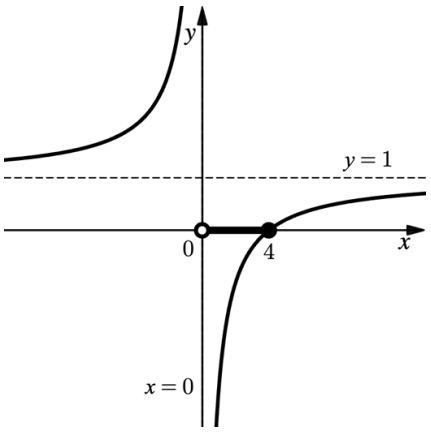


1 b)
i)



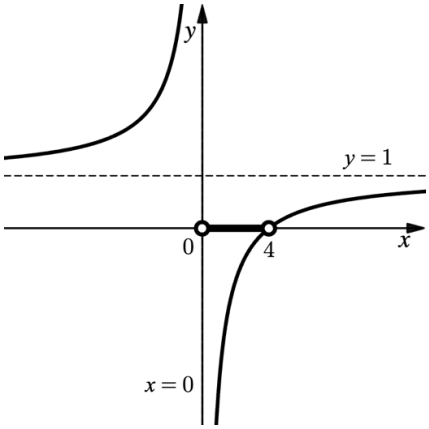
$$\{x : x \in \mathbb{R}, x < 0\} \cup \{x : x \in \mathbb{R}, x > 4\}$$

ii)



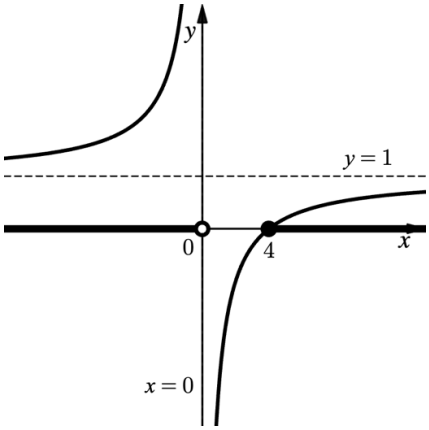
$$\{x : x \in \mathbb{R}, 0 < x \leq 4\}$$

1 c)
i)



$$x \in (0, 4)$$

ii)



$$x \in (-\infty, 0) \cup [4, \infty)$$

2 a)

$$\frac{1}{x} > 0$$

Only critical value is $x = 0$

$x < 0$	$x > 0$
$-$	$+$

$$\{x : x \in \mathbb{R}, x > 0\}$$

2 b)

$$\frac{1}{x-3} < 0$$

Only critical value is $x = 3$

$x < 3$	$x > 3$
$-$	$+$

$$\{x : x \in \mathbb{R}, x < 3\}$$

2 c)

$$\frac{2}{x+1} \geq 0$$

Only critical value is $x = -1$

$x < -1$	$x > -1$
$-$	$+$

$$\{x : x \in \mathbb{R}, x > -1\}$$

2 d)

$$\frac{5}{2x+3} \leq 0$$

Only critical value is $x = -\frac{3}{2}$

$x < -\frac{3}{2}$	$x > -\frac{3}{2}$
$-$	$+$

$$\{x : x \in \mathbb{R}, x < -\frac{3}{2}\}$$

2 e)

$$\frac{8}{4-7x} > 0$$

Only critical value is $x = \frac{4}{7}$

$x < \frac{4}{7}$	$x > \frac{4}{7}$
$+$	$-$

$$\{x : x \in \mathbb{R}, x < \frac{4}{7}\}$$

3 a)

$$\frac{1}{x} > 2 \Rightarrow \frac{1}{x} \times x^2 > 2 \times x^2 \Rightarrow x > 2x^2 \Rightarrow x - 2x^2 > 0$$
$$\Rightarrow (1 - 2x)x > 0$$

Critical values at $x = 0$ and $x = \frac{1}{2}$

$x < 0$	$0 < x < \frac{1}{2}$	$x > \frac{1}{2}$
$-$	$+$	$-$

$$x \in (0, \frac{1}{2})$$

3 b)

$$\frac{1}{x-2} < 3$$
$$\Rightarrow \frac{1}{x-2} \times (x-2)^2 < 3 \times (x-2)^2$$
$$\Rightarrow x - 2 < 3(x-2)^2$$
$$\Rightarrow (x-2) - 3(x-2)^2 < 0$$
$$\Rightarrow (1 - 3(x-2))(x-2) < 0$$
$$\Rightarrow (7 - 3x)(x-2) < 0$$

Critical values at $x = 2$ and $x = \frac{7}{3}$

$x < 2$	$2 < x < \frac{7}{3}$	$x > \frac{7}{3}$
$-$	$+$	$-$

$$x \in (-\infty, 2) \cup (\frac{7}{3}, \infty)$$

3 c)

$$\frac{3}{x+4} \geq -5$$
$$\Rightarrow \frac{3}{x+4} \times (x+4)^2 \geq -5 \times (x+4)^2$$
$$\Rightarrow 3(x+4) \geq -5(x+4)^2$$
$$\Rightarrow 3(x+4) + 5(x+4)^2 \geq 0$$
$$\Rightarrow (3 + 5(x+4))(x+4) \geq 0$$
$$\Rightarrow (5x + 23)(x+4) \geq 0$$

Critical values at $x = -\frac{23}{5}$ and $x = -4$

$x < -\frac{23}{5}$	$-\frac{23}{5} < x < -4$	$x > -4$
$+$	$-$	$+$

$$x \in (-\infty, -\frac{23}{5}] \cup (-4, \infty)$$

3 d)

$$\frac{4}{3-2x} \leq 7$$
$$\Rightarrow \frac{4}{3-2x} \times (3-2x)^2 \leq 7 \times (3-2x)^2$$
$$\Rightarrow 4(3-2x) \leq 7(3-2x)^2$$
$$\Rightarrow 4(3-2x) - 7(3-2x)^2 \leq 0$$
$$\Rightarrow (4-7(3-2x))(3-2x) \leq 0$$
$$\Rightarrow (14x-17)(3-2x) \leq 0$$

Critical values at $x = \frac{17}{14}$ and $x = \frac{3}{2}$

$x < \frac{17}{14}$	$\frac{17}{14} < x < \frac{3}{2}$	$x > \frac{3}{2}$
$-$	$+$	$-$

$$x \in (-\infty, \frac{17}{14}] \cup (\frac{3}{2}, \infty)$$

3 e)

$$\frac{10}{6-3x} > 9$$
$$\Rightarrow \frac{10}{6-3x} \times (6-3x)^2 > 9 \times (6-3x)^2$$
$$\Rightarrow 10(6-3x) > 9(6-3x)^2$$
$$\Rightarrow 10(6-3x) - 9(6-3x)^2 > 0$$
$$\Rightarrow (10-9(6-3x))(6-3x) > 0$$
$$\Rightarrow (27x-44)(6-3x) > 0$$
$$\Rightarrow (27x-44)(2-x) > 0$$

Critical values at $x = \frac{44}{27}$ and $x = 2$

$x < \frac{44}{27}$	$\frac{44}{27} < x < 2$	$x > 2$
$-$	$+$	$-$

$$x \in (\frac{44}{27}, 2)$$

4 a)

The error was made when Francesca multiplied both sides of the inequality by $(x-3)$. As $(x-3)$ could be negative, this would change the direction of the inequality. Francesca should instead multiply both sides of the inequality by $(x-3)^2$ at this stage.

4 b)

$$\frac{5}{x-3} \geq 9-x$$
$$\Rightarrow \frac{5}{x-3} \times (x-3)^2 \geq (9-x) \times (x-3)^2$$
$$\Rightarrow 5(x-3) \geq (9-x)(x-3)^2$$
$$\Rightarrow 5(x-3) - (9-x)(x-3)^2 \geq 0$$
$$\Rightarrow (5-(9-x)(x-3))(x-3) \geq 0$$
$$\Rightarrow (5-(-x^2+12x-27))(x-3) \geq 0$$
$$\Rightarrow (x^2-12x+32)(x-3) \geq 0$$
$$\Rightarrow (x-4)(x-8)(x-3) \geq 0$$

Critical values at $x=3$, $x=4$ and $x=8$

$x<3$	$3<x<4$	$4<x<8$	$x>8$
$-$	$+$	$-$	$+$

$$x \in (3,4] \cup [8,\infty)$$

5 a)

$$\frac{x+1}{x} > 0$$

Critical values at $x=-1$ and $x=0$

$x<-1$	$-1<x<0$	$x>0$
$+$	$-$	$+$

$$\{x : x \in \mathbb{R}, x < -1\} \cup \{x : x \in \mathbb{R}, x > 0\}$$

5 b)

$$\frac{x-1}{2x} < 0$$

Critical values at $x=0$ and $x=1$

$x<0$	$0<x<1$	$x>1$
$+$	$-$	$+$

$$\{x : x \in \mathbb{R}, 0 < x < 1\}$$

5 c)

$$\frac{2x-3}{4x} \geq 0$$

Critical values at $x=0$ and $x=\frac{3}{2}$

$x<0$	$0<x<\frac{3}{2}$	$x>\frac{3}{2}$
$+$	$-$	$+$

$$\{x : x \in \mathbb{R}, x < 0\} \cup \{x : x \in \mathbb{R}, x \geq \frac{3}{2}\}$$

5 d)

$$\frac{5-3x}{6x} \leq 0$$

Critical values at $x = 0$ and $x = \frac{5}{3}$

$x < 0$	$0 < x < \frac{5}{3}$	$x > \frac{5}{3}$
$-$	$+$	$-$

$$\{x : x \in \mathbb{R}, x < 0\} \cup \{x : x \in \mathbb{R}, x \geq \frac{5}{3}\}$$

6 a)

$$\frac{x+2}{x} > 1$$
$$\Rightarrow \frac{x+2}{x} \times x^2 > 1 \times x^2$$
$$\Rightarrow (x+2)x > x^2$$
$$\Rightarrow (x+2)x - x^2 > 0$$
$$\Rightarrow ((x+2) - x)x > 0$$
$$\Rightarrow 2x > 0$$
$$\Rightarrow x > 0$$

$$x \in (0, \infty)$$

6 b)

$$\frac{x-4}{x} < 1$$
$$\Rightarrow \frac{x-4}{x} \times x^2 < 1 \times x^2$$
$$\Rightarrow (x-4)x < x^2$$
$$\Rightarrow (x-4)x - x^2 < 0$$
$$\Rightarrow ((x-4) - x)x < 0$$
$$\Rightarrow -4x < 0$$
$$\Rightarrow x > 0$$

$$x \in (0, \infty)$$

6 c)

$$\frac{2x+1}{2x} \geq 2$$
$$\Rightarrow \frac{2x+1}{2x} \times x^2 \geq 2 \times x^2$$
$$\Rightarrow (2x+1)x \geq 4x^2$$
$$\Rightarrow (2x+1)x - 4x^2 \geq 0$$
$$\Rightarrow ((2x+1) - 4x)x \geq 0$$
$$\Rightarrow (1-2x)x \geq 0$$

Critical values at $x = 0$ and $x = \frac{1}{2}$

$x < 0$	$0 < x < \frac{1}{2}$	$x > \frac{1}{2}$
$-$	$+$	$-$

$$x \in (0, \frac{1}{2}]$$

6 d)

$$\frac{5-3x}{x} \leq -4$$
$$\Rightarrow \frac{5-3x}{x} \times x^2 \leq -4 \times x^2$$
$$\Rightarrow (5-3x)x \leq -4x^2$$
$$\Rightarrow (5-3x)x + 4x^2 \leq 0$$
$$\Rightarrow ((5-3x) + 4x)x \leq 0$$
$$\Rightarrow (5+x)x \leq 0$$

Critical values at $x = -5$ and $x = 0$

$x < -5$	$-5 < x < 0$	$x > 0$
$+$	$-$	$+$

$$x \in [-5, 0)$$

7 a)

$$\frac{x}{x+1} > 0$$

Critical values at $x = -1$ and $x = 0$

$x < -1$	$-1 < x < 0$	$x > 0$
$+$	$-$	$+$

$$\{x : x \in \mathbb{R}, x < -1\} \cup \{x : x \in \mathbb{R}, x > 0\}$$

7 b)

$$\frac{2x}{x-3} < 0$$

Critical values at $x = 0$ and $x = 3$

$x < 0$	$0 < x < 3$	$x > 3$
$+$	$-$	$+$

$$\{x : x \in \mathbb{R}, 0 < x < 3\}$$

7 c)

$$\frac{4x}{2x+3} \geq 0$$

Critical values at $x = -\frac{3}{2}$ and $x = 0$

$x < -\frac{3}{2}$	$-\frac{3}{2} < x < 0$	$x > 0$
$+$	$-$	$+$

$$\{x : x \in \mathbb{R}, x < -\frac{3}{2}\} \cup \{x : x \in \mathbb{R}, x \geq 0\}$$

7 d)

$$\frac{3x}{3-x} \leq 0$$

Critical values at $x = 0$ and $x = 3$

$x < 0$	$0 < x < 3$	$x > 3$
$-$	$+$	$-$

$$\{x : x \in \mathbb{R}, x \leq 0\} \cup \{x : x \in \mathbb{R}, x > 3\}$$

8 a)

$$\frac{x-3}{x+1} > 0$$

Critical values at $x = -1$ and $x = 3$

$x < -1$	$-1 < x < 3$	$x > 3$
$+$	$-$	$+$

$$x \in (-\infty, -1) \cup (3, \infty)$$

8 b)

$$\frac{x+4}{x-5} < 0$$

Critical values at $x = -4$ and $x = 5$

$x < -4$	$-4 < x < 5$	$x > 5$
$+$	$-$	$+$

$$x \in (-4, 5)$$

8 c)

$$\frac{2x+1}{x+3} \geq 0$$

Critical values at $x = -3$ and $x = -\frac{1}{2}$

$x < -3$	$-3 < x < -\frac{1}{2}$	$x > -\frac{1}{2}$
$+$	$-$	$+$

$$x \in (-\infty, -3) \cup [-\frac{1}{2}, \infty)$$

8 d)

$$\frac{5-x}{2x+5} \leq 0$$

Critical values at $x = -\frac{5}{2}$ and $x = 5$

$x < -\frac{5}{2}$	$-\frac{5}{2} < x < 5$	$x > 5$
$-$	$+$	$-$

$$x \in (-\infty, -\frac{5}{2}) \cup [5, \infty)$$

9 a)

$$\frac{x+3}{x-2} > 1$$
$$\Rightarrow \frac{x+3}{x-2} \times (x-2)^2 > 1 \times (x-2)^2$$
$$\Rightarrow (x+3)(x-2) > (x-2)^2$$
$$\Rightarrow (x+3)(x-2) - (x-2)^2 > 0$$
$$\Rightarrow ((x+3) - (x-2))(x-2) > 0$$
$$\Rightarrow 5(x-2) > 0$$
$$\Rightarrow x - 2 > 0$$
$$\Rightarrow x > 2$$

$$\{x : x \in \mathbb{R}, x > 2\}$$

9 b)

$$\frac{x-6}{x+1} < 1$$
$$\Rightarrow \frac{x-6}{x+1} \times (x+1)^2 < 1 \times (x+1)^2$$
$$\Rightarrow (x-6)(x+1) < (x+1)^2$$
$$\Rightarrow (x-6)(x+1) - (x+1)^2 < 0$$
$$\Rightarrow ((x-6) - (x+1))(x+1) < 0$$
$$\Rightarrow -7(x+1) < 0$$
$$\Rightarrow x+1 > 0$$
$$\Rightarrow x > -1$$

$$\{x : x \in \mathbb{R}, x > -1\}$$

9 c)

$$\frac{x+4}{2x-1} \geq -3$$
$$\Rightarrow \frac{x+4}{2x-1} \times (2x-1)^2 \geq -3 \times (2x-1)^2$$
$$\Rightarrow (x+4)(2x-1) \geq -3(2x-1)^2$$
$$\Rightarrow (x+4)(2x-1) + 3(2x-1)^2 \geq 0$$
$$\Rightarrow ((x+4) + 3(2x-1))(2x-1) \geq 0$$
$$\Rightarrow (7x+1)(2x-1) \geq 0$$

Critical values at $x = -\frac{1}{7}$ and $x = \frac{1}{2}$

$x < -\frac{1}{7}$	$-\frac{1}{7} < x < \frac{1}{2}$	$x > \frac{1}{2}$
$+$	$-$	$+$

$$\{x : x \in \mathbb{R}, x \leq -\frac{1}{7}\} \cup \{x : x \in \mathbb{R}, x > \frac{1}{2}\}$$

9 d)

$$\frac{7-x}{x-8} \leq 5$$
$$\Rightarrow \frac{7-x}{x-8} \times (x-8)^2 \leq 5 \times (x-8)^2$$
$$\Rightarrow (7-x)(x-8) \leq 5(x-8)^2$$
$$\Rightarrow (7-x)(x-8) - 5(x-8)^2 \leq 0$$
$$\Rightarrow ((7-x) - 5(x-8))(x-8) \leq 0$$
$$\Rightarrow (47-6x)(x-8) \leq 0$$

Critical values at $x = \frac{47}{6}$ and $x = 8$

$x < \frac{47}{6}$	$\frac{47}{6} < x < 8$	$x > 8$
$-$	$+$	$-$

$$\{x : x \in \mathbb{R}, x \leq \frac{47}{6}\} \cup \{x : x \in \mathbb{R}, x > 8\}$$

10 a)

First solve $\frac{2x+5}{x-5} > 0$ Critical values at $x = -\frac{5}{2}$ and $x = 5$

$x < -\frac{5}{2}$	$-\frac{5}{2} < x < 5$	$x > 5$
$+$	$-$	$+$

$$\{x : x \in \mathbb{R}, x < -\frac{5}{2}\} \cup \{x : x \in \mathbb{R}, x > 5\}$$

Now solve $\frac{2x+5}{x-5} < 1$

$$\Rightarrow \frac{2x+5}{x-5} \times (x-5)^2 < 1 \times (x-5)^2$$
$$\Rightarrow (2x+5)(x-5) < (x-5)^2$$
$$\Rightarrow (2x+5)(x-5) - (x-5)^2 < 0$$
$$\Rightarrow ((2x+5) - (x-5))(x-5) < 0$$
$$\Rightarrow (x+10)(x-5) < 0$$

Critical values at $x = -10$ and $x = 5$

$x < -10$	$-10 < x < 5$	$x > 5$
$+$	$-$	$+$

$$\{x : x \in \mathbb{R}, -10 < x < 5\}$$

Both inequalities are true when $\{x : x \in \mathbb{R}, -10 < x < -\frac{5}{2}\}$

10 b)
First solve $\frac{3x+1}{2-x} > -1$

$$\Rightarrow \frac{3x+1}{2-x} \times (2-x)^2 > -1 \times (2-x)^2$$
$$\Rightarrow (3x+1)(2-x) > -(2-x)^2$$
$$\Rightarrow (3x+1)(2-x) + (2-x)^2 > 0$$
$$\Rightarrow ((3x+1) + (2-x))(2-x) > 0$$
$$\Rightarrow (2x+3)(2-x) > 0$$

Critical values at $x = -\frac{3}{2}$ and $x = 2$

$x < -\frac{3}{2}$	$-\frac{3}{2} < x < 2$	$x > 2$
$-$	$+$	$-$

$$\{x : x \in \mathbb{R}, -\frac{3}{2} < x < 2\}$$

Now solve $\frac{3x+1}{2-x} < 1$

$$\Rightarrow \frac{3x+1}{2-x} \times (2-x)^2 < 1 \times (2-x)^2$$
$$\Rightarrow (3x+1)(2-x) < (2-x)^2$$
$$\Rightarrow (3x+1)(2-x) - (2-x)^2 < 0$$
$$\Rightarrow ((3x+1) - (2-x))(2-x) < 0$$
$$\Rightarrow (4x-1)(2-x) < 0$$

Critical values at $x = \frac{1}{4}$ and $x = 2$

$x < \frac{1}{4}$	$\frac{1}{4} < x < 2$	$x > 2$
$-$	$+$	$-$

$$\{x : x \in \mathbb{R}, x < \frac{1}{4}\} \cup \{x : x \in \mathbb{R}, x > 2\}$$

Both inequalities are true when $\{x : x \in \mathbb{R}, -\frac{3}{2} < x < \frac{1}{4}\}$

10 c)
First solve $\frac{5-x}{4x+1} \geq -2$

$$\begin{aligned}
\Rightarrow \frac{5-x}{4x+1} \times (4x+1)^2 &\geq -2 \times (4x+1)^2 \\
\Rightarrow (5-x)(4x+1) &\geq -2(4x+1)^2 \\
\Rightarrow (5-x)(4x+1) + 2(4x+1)^2 &\geq 0 \\
\Rightarrow ((5-x) + 2(4x+1))(4x+1) &\geq 0 \\
\Rightarrow (7x+7)(4x+1) &\geq 0 \\
\Rightarrow (x+1)(4x+1) &\geq 0
\end{aligned}$$

Critical values at $x = -1$ and $x = -\frac{1}{4}$

$x < -1$	$-1 < x < -\frac{1}{4}$	$x > -\frac{1}{4}$
$+$	$-$	$+$

$$\{x : x \in \mathbb{R}, x \leq -1\} \cup \{x : x \in \mathbb{R}, x \geq -\frac{1}{4}\}$$

Now solve $\frac{5-x}{4x+1} \leq 2$

$$\begin{aligned}
\Rightarrow \frac{5-x}{4x+1} \times (4x+1)^2 &\leq 2 \times (4x+1)^2 \\
\Rightarrow (5-x)(4x+1) &\leq 2(4x+1)^2 \\
\Rightarrow (5-x)(4x+1) - 2(4x+1)^2 &\leq 0 \\
\Rightarrow ((5-x) - 2(4x+1))(4x+1) &\leq 0 \\
\Rightarrow (3-9x)(4x+1) &\leq 0 \\
\Rightarrow (1-3x)(4x+1) &\leq 0
\end{aligned}$$

Critical values at $x = -\frac{1}{4}$ and $x = \frac{1}{3}$

$x < -\frac{1}{4}$	$-\frac{1}{4} < x < \frac{1}{3}$	$x > \frac{1}{3}$
$-$	$+$	$-$

$$\{x : x \in \mathbb{R}, x < -\frac{1}{4}\} \cup \{x : x \in \mathbb{R}, x \geq \frac{1}{3}\}$$

Both inequalities are true when $\{x : x \in \mathbb{R}, x \leq -1\} \cup \{x : x \in \mathbb{R}, x \geq \frac{1}{3}\}$

11 a)

$$\begin{aligned}
\frac{2}{x} &< 2x-3 \\
\Rightarrow \frac{2}{x}\times x^2 &< (2x-3)\times x^2 \\
\Rightarrow 2x &< (2x-3)x^2 \\
\Rightarrow 2x-(2x-3)x^2 &< 0 \\
\Rightarrow (2-(2x-3)x)x &< 0 \\
\Rightarrow (2+3x-2x^2)x &< 0 \\
\Rightarrow (2x+1)(2-x)x &< 0
\end{aligned}$$

Critical values at $x=-\frac{1}{2}$, $x=0$ and $x=2$

$x<-\frac{1}{2}$	$-\frac{1}{2}<x<0$	$0<x<2$	$x>2$
$+$	$-$	$+$	$-$

$$x\in(-\frac{1}{2},0)\cup(2,\infty)$$

11 b)

$$\begin{aligned}
\frac{5}{2x+1} &\leq 3x-2 \\
\Rightarrow \frac{5}{2x+1}\times(2x+1)^2 &\leq (3x-2)\times(2x+1)^2 \\
\Rightarrow 5(2x+1) &\leq (3x-2)(2x+1)^2 \\
\Rightarrow 5(2x+1)-(3x-2)(2x+1)^2 &\leq 0 \\
\Rightarrow (5-(3x-2)(2x+1))(2x+1) &\leq 0 \\
\Rightarrow (-6x^2+x+7)(2x+1) &\leq 0 \\
\Rightarrow (7-6x)(x+1)(2x+1) &\leq 0
\end{aligned}$$

Critical values at $x=-1$, $x=-\frac{1}{2}$ and $x=\frac{7}{6}$

$x<-1$	$-1<x<-\frac{1}{2}$	$-\frac{1}{2}<x<\frac{7}{6}$	$x>\frac{7}{6}$
$+$	$-$	$+$	$-$

$$x\in[-1,-\frac{1}{2})\cup[\frac{7}{6},\infty)$$

11 c)

$$\begin{aligned}
&\frac{x+3}{x} > x-1 \\
\Rightarrow &\frac{x+3}{x} \times x^2 > (x-1) \times x^2 \\
\Rightarrow &(x+3)x > (x-1)x^2 \\
\Rightarrow &(x+3)x - (x-1)x^2 > 0 \\
\Rightarrow &((x+3) - (x-1)x)x > 0 \\
\Rightarrow &(3+2x-x^2)x > 0 \\
\Rightarrow &(x+1)(3-x)x > 0
\end{aligned}$$

Critical values at $x = -1$, $x = 0$ and $x = 3$

$x < -1$	$-1 < x < 0$	$0 < x < 3$	$x > 3$
$+$	$-$	$+$	$-$

$$x \in (-\infty, -1) \cup (0, 3)$$

11 d)

$$\begin{aligned}
&\frac{x-5}{2x} < 2-x \\
\Rightarrow &\frac{x-5}{2x} \times x^2 < (2-x) \times x^2 \\
\Rightarrow &(x-5)x < 2(2-x)x^2 \\
\Rightarrow &(x-5)x - 2(2-x)x^2 < 0 \\
\Rightarrow &((x-5) - 2(2-x)x)x < 0 \\
\Rightarrow &(2x^2 - 3x - 5)x < 0 \\
\Rightarrow &(2x-5)(x+1)x < 0
\end{aligned}$$

Critical values at $x = -1$, $x = 0$ and $x = \frac{5}{2}$

$x < -1$	$-1 < x < 0$	$0 < x < \frac{5}{2}$	$x > \frac{5}{2}$
$-$	$+$	$-$	$+$

$$x \in (-\infty, -1) \cup (0, \frac{5}{2})$$

11 e)

$$\frac{x}{x-4} \geq 2x+1$$
$$\Rightarrow \frac{x}{x-4} \times (x-4)^2 \geq (2x+1) \times (x-4)^2$$
$$\Rightarrow x(x-4) \geq (2x+1)(x-4)^2$$
$$\Rightarrow x(x-4) - (2x+1)(x-4)^2 \geq 0$$
$$\Rightarrow (x-(2x+1)(x-4))(x-4) \geq 0$$
$$\Rightarrow (4+8x-2x^2)(x-4) \geq 0$$
$$\Rightarrow (2+4x-x^2)(x-4) \geq 0$$

Critical values at $x = 2-\sqrt{6}$, $x = 4$ and $x = 2+\sqrt{6}$

$x < 2-\sqrt{6}$	$2-\sqrt{6} < x < 4$	$4 < x < 2+\sqrt{6}$	$x > 2+\sqrt{6}$
$+$	$-$	$+$	$-$

$$x \in (-\infty, 2-\sqrt{6}] \cup (4, 2+\sqrt{6}]$$

11 f)

$$\frac{3x}{5-2x} \leq 2x-3$$
$$\Rightarrow \frac{3x}{5-2x} \times (5-2x)^2 \leq (2x-3) \times (5-2x)^2$$
$$\Rightarrow 3x(5-2x) \leq (2x-3)(5-2x)^2$$
$$\Rightarrow 3x(5-2x) - (2x-3)(5-2x)^2 \leq 0$$
$$\Rightarrow (3x-(2x-3)(5-2x))(5-2x) \leq 0$$
$$\Rightarrow (4x^2-13x+15)(5-2x) \leq 0$$

For $4x^2-13x+15$, $b^2-4ac = (-13)^2-4(4)(15) = -71 < 0$ so $4x^2-13x+15 > 0$ for all real values of x

$$\Rightarrow 5-2x \leq 0$$
$$\Rightarrow -2x \leq -5$$
$$\Rightarrow x \geq \frac{5}{2}$$

However, $x \neq \frac{5}{2}$ so $x \in (\frac{5}{2}, \infty)$

11 g)

$$\frac{x+2}{x+6} > x-5$$
$$\Rightarrow \frac{x+2}{x+6} \times (x+6)^2 > (x-5) \times (x+6)^2$$
$$\Rightarrow (x+2)(x+6) > (x-5)(x+6)^2$$
$$\Rightarrow (x+2)(x+6) - (x-5)(x+6)^2 > 0$$
$$\Rightarrow ((x+2) - (x-5)(x+6))(x+6) > 0$$
$$\Rightarrow (32 - x^2)(x+6) > 0$$

Critical values at $x = -6$, $x = -4\sqrt{2}$ and $x = 4\sqrt{2}$

$x < -6$	$-6 < x < -4\sqrt{2}$	$-4\sqrt{2} < x < 4\sqrt{2}$	$x > 4\sqrt{2}$
$+$	$-$	$+$	$-$

$$x \in (-\infty, -6) \cup (-4\sqrt{2}, 4\sqrt{2})$$

11 h)

$$\frac{4-x}{6-5x} \geq 2x-1$$
$$\Rightarrow \frac{4-x}{6-5x} \times (6-5x)^2 \geq (2x-1) \times (6-5x)^2$$
$$\Rightarrow (4-x)(6-5x) \geq (2x-1)(6-5x)^2$$
$$\Rightarrow (4-x)(6-5x) - (2x-1)(6-5x)^2 \geq 0$$
$$\Rightarrow ((4-x) - (2x-1)(6-5x))(6-5x) \geq 0$$
$$\Rightarrow (10x^2 - 18x + 10)(6-5x) \geq 0$$
$$\Rightarrow (5x^2 - 9x + 5)(6-5x) \geq 0$$

For $5x^2 - 9x + 5$, $b^2 - 4ac = (-9)^2 - 4(5)(5) = -19 < 0$ so $5x^2 - 9x + 5 > 0$ for all real values of x

$$\Rightarrow 6 - 5x \geq 0$$
$$\Rightarrow -5x \geq -6$$
$$\Rightarrow x \leq \frac{6}{5}$$

However, $x \neq \frac{6}{5}$ so $x \in (-\infty, \frac{6}{5})$

11 i)

$$\frac{3-2x}{2-3x} \leq 6-x$$
$$\Rightarrow \frac{3-2x}{2-3x} \times (2-3x)^2 \leq (6-x) \times (2-3x)^2$$
$$\Rightarrow (3-2x)(2-3x) \leq (6-x)(2-3x)^2$$
$$\Rightarrow (3-2x)(2-3x) - (6-x)(2-3x)^2 \leq 0$$
$$\Rightarrow ((3-2x) - (6-x)(2-3x))(2-3x) \leq 0$$
$$\Rightarrow (-3x^2+18x-9)(2-3x) \leq 0$$
$$\Rightarrow (-x^2+6x-3)(2-3x) \leq 0$$

Critical values at $x = 3-\sqrt{6}$, $x = \frac{2}{3}$ and $x = 3+\sqrt{6}$

$x < 3-\sqrt{6}$	$3-\sqrt{6} < x < \frac{2}{3}$	$\frac{2}{3} < x < 3+\sqrt{6}$	$x > 3+\sqrt{6}$
$-$	$+$	$-$	$+$

$$x \in (-\infty, 3-\sqrt{6}] \cup (\frac{2}{3}, 3+\sqrt{6}]$$

12 a)

$$\frac{(x-1)(x-2)}{x} > 0$$

Critical values at $x = 0$, $x = 1$ and $x = 2$

$x < 0$	$0 < x < 1$	$1 < x < 2$	$x > 2$
$-$	$+$	$-$	$+$

$$\{x : x \in \mathbb{R}, 0 < x < 1\} \cup \{x : x \in \mathbb{R}, x > 2\}$$

12 b)

$$\frac{(x+3)(x-5)}{x} < 0$$

Critical values at $x = -3$, $x = 0$ and $x = 5$

$x < -3$	$-3 < x < 0$	$0 < x < 5$	$x > 5$
$-$	$+$	$-$	$+$

$$\{x : x \in \mathbb{R}, x < -3\} \cup \{x : x \in \mathbb{R}, 0 < x < 5\}$$

12 c)

$$\frac{(2x-1)(3x+4)}{2x} \leq 0$$

Critical values at $x = -\frac{4}{3}$, $x = 0$ and $x = \frac{1}{2}$

$x < -\frac{4}{3}$	$-\frac{4}{3} < x < 0$	$0 < x < \frac{1}{2}$	$x > \frac{1}{2}$
$-$	$+$	$-$	$+$

$$\{x : x \in \mathbb{R}, x \leq -\frac{4}{3}\} \cup \{x : x \in \mathbb{R}, 0 < x \leq \frac{1}{2}\}$$

12 d)

$$\frac{(3-x)(1+2x)}{4x} \geq 0$$

Critical values at $x = -\frac{1}{2}$, $x = 0$ and $x = 3$

$x < -\frac{1}{2}$	$-\frac{1}{2} < x < 0$	$0 < x < 3$	$x > 3$
$+$	$-$	$+$	$-$

$$\{x : x \in \mathbb{R}, x \leq -\frac{1}{2}\} \cup \{x : x \in \mathbb{R}, 0 < x \leq 3\}$$

13 a)

$$\frac{x-3}{(x+1)(x+2)} > 0$$

Critical values at $x = -2$, $x = -1$ and $x = 3$

$x < -2$	$-2 < x < -1$	$-1 < x < 3$	$x > 3$
$-$	$+$	$-$	$+$

$$x \in (-2, -1) \cup (3, \infty)$$

13 b)

$$\frac{x+4}{(x-2)(x-5)} < 0$$

Critical values at $x = -4$, $x = 2$ and $x = 5$

$x < -4$	$-4 < x < 2$	$2 < x < 5$	$x > 5$
$-$	$+$	$-$	$+$

$$x \in (-\infty, -4) \cup (2, 5)$$

13 c)

$$\frac{2x+1}{(x+3)(x-4)} \geq 0$$

Critical values at $x = -3$, $x = -\frac{1}{2}$ and $x = 4$

$x < -3$	$-3 < x < -\frac{1}{2}$	$-\frac{1}{2} < x < 4$	$x > 4$
$-$	$+$	$-$	$+$

$$x \in (-3, -\frac{1}{2}] \cup (4, \infty)$$

13 d)

$$\frac{3-5x}{(2x+1)(x+3)} \leq 0$$

Critical values at $x = -3$, $x = -\frac{1}{2}$ and $x = \frac{3}{5}$

$x < -3$	$-3 < x < -\frac{1}{2}$	$-\frac{1}{2} < x < \frac{3}{5}$	$x > \frac{3}{5}$
$+$	$-$	$+$	$-$

$$x \in (-3, -\frac{1}{2}) \cup [\frac{3}{5}, \infty)$$

14 a)

$$\frac{(x-4)(x-3)}{x+2} > 0$$

Critical values at $x = -2$, $x = 3$ and $x = 4$

$x < -2$	$-2 < x < 3$	$3 < x < 4$	$x > 4$
$-$	$+$	$-$	$+$

$$\{x : x \in \mathbb{R}, -2 < x < 3\} \cup \{x : x \in \mathbb{R}, x > 4\}$$

14 b)

$$\frac{(x+1)(x+5)}{x-3} < 0$$

Critical values at $x = -5$, $x = -1$ and $x = 3$

$x < -5$	$-5 < x < -1$	$-1 < x < 3$	$x > 3$
$-$	$+$	$-$	$+$

$$\{x : x \in \mathbb{R}, x < -5\} \cup \{x : x \in \mathbb{R}, -1 < x < 3\}$$

14 c)

$$\frac{(2x-3)(x+2)}{x-1} \geq 0$$

Critical values at $x = -2$, $x = 1$ and $x = \frac{3}{2}$

$x < -2$	$-2 < x < 1$	$1 < x < \frac{3}{2}$	$x > \frac{3}{2}$
$-$	$+$	$-$	$+$

$$\{x : x \in \mathbb{R}, -2 \leq x < 1\} \cup \{x : x \in \mathbb{R}, x \geq \frac{3}{2}\}$$

14 d)

$$\frac{(4-x)(x-6)}{2x+3} \leq 0$$

Critical values at $x = -\frac{3}{2}$, $x = 4$ and $x = 6$

$x < -\frac{3}{2}$	$-\frac{3}{2} < x < 4$	$4 < x < 6$	$x > 6$
$+$	$-$	$+$	$-$

$$\{x : x \in \mathbb{R}, -\frac{3}{2} < x \leq 4\} \cup \{x : x \in \mathbb{R}, x \geq 6\}$$

15 a)

$$\begin{aligned}
\frac{x-3}{x+5} &> x + 12 \\
\Rightarrow \frac{x-3}{x+5} - (x+12) &> 0 \\
\Rightarrow \frac{x-3}{x+5} - \frac{(x+12)(x+5)}{x+5} &> 0 \\
\Rightarrow \frac{(x-3)-(x+12)(x+5)}{x+5} &> 0 \\
\Rightarrow \frac{-x^2+16x+63}{x+5} &> 0 \\
\Rightarrow -\frac{(x+9)(x+7)}{x+5} &> 0 \\
\Rightarrow \frac{(x+9)(x+7)}{x+5} &< 0 \\
\therefore k &= 7
\end{aligned}$$

15 b)

Critical values at $x = -9$, $x = -7$ and $x = -5$

$x < -9$	$-9 < x < -7$	$-7 < x < -5$	$x > -5$
$-$	$+$	$-$	$+$

$$x \in (-\infty, -9) \cup (-7, -5)$$

16 a)

$$\frac{(3x-1)(x+1)}{x+2} \geq 12$$
$$\Rightarrow \frac{(3x-1)(x+1)}{x+2} \times (x+2)^2 \geq 12 \times (x+2)^2$$
$$\Rightarrow (3x-1)(x+1)(x+2) \geq 12(x+2)^2$$
$$\Rightarrow (3x-1)(x+1)(x+2) - 12(x+2)^2 \geq 0$$
$$\Rightarrow ((3x-1)(x+1) - 12(x+2))(x+2) \geq 0$$
$$\Rightarrow (3x^2 - 10x - 25)(x+2) \geq 0$$
$$\Rightarrow (3x+5)(x-5)(x+2) \geq 0$$

Critical values at $x = -2$, $x = -\frac{5}{3}$ and $x = 5$

$x < -2$	$-2 < x < -\frac{5}{3}$	$-\frac{5}{3} < x < 5$	$x > 5$
$-$	$+$	$-$	$+$

$$\{x : x \in \mathbb{R}, -2 < x \leq -\frac{5}{3}\} \cup \{x : x \in \mathbb{R}, x \geq 5\}$$

16 b)

$$\frac{(8-x)(2x+5)}{3x+1} \leq 4$$
$$\Rightarrow \frac{(8-x)(2x+5)}{3x+1} \times (3x+1)^2 \leq 4 \times (3x+1)^2$$
$$\Rightarrow (8-x)(2x+5)(3x+1) \leq 4(3x+1)^2$$
$$\Rightarrow (8-x)(2x+5)(3x+1) - 4(3x+1)^2 \leq 0$$
$$\Rightarrow ((8-x)(2x+5) - 4(3x+1))(3x+1) \leq 0$$
$$\Rightarrow (-2x^2 - x + 36)(3x+1) \leq 0$$
$$\Rightarrow (2x+9)(4-x)(3x+1) \leq 0$$

Critical values at $x = -\frac{9}{2}$, $x = -\frac{1}{3}$ and $x = 4$

$x < -\frac{9}{2}$	$-\frac{9}{2} < x < -\frac{1}{3}$	$-\frac{1}{3} < x < 4$	$x > 4$
$+$	$-$	$+$	$-$

$$\{x : x \in \mathbb{R}, -\frac{9}{2} \leq x < -\frac{1}{3}\} \cup \{x : x \in \mathbb{R}, x \geq 4\}$$

16 c)

$$\frac{(2-5x)(4-x)}{6-x} < -4$$
$$\Rightarrow \frac{(2-5x)(4-x)}{6-x} \times (6-x)^2 < -4 \times (6-x)^2$$
$$\Rightarrow (2-5x)(4-x)(6-x) < -4(6-x)^2$$
$$\Rightarrow (2-5x)(4-x)(6-x) + 4(6-x)^2 < 0$$
$$\Rightarrow ((2-5x)(4-x) + 4(6-x))(6-x) < 0$$
$$\Rightarrow (5x^2 - 26x + 32)(6-x) < 0$$
$$\Rightarrow (5x-16)(x-2)(6-x) < 0$$

Critical values at $x = 2$, $x = \frac{16}{5}$ and $x = 6$

$x < -2$	$2 < x < \frac{16}{5}$	$\frac{16}{5} < x < 6$	$x > 6$
$+$	$-$	$+$	$-$

$$\{x : x \in \mathbb{R}, 2 < x < \frac{16}{5}\} \cup \{x : x \in \mathbb{R}, x > 6\}$$

17 a)

$$\frac{x^2-9}{5x-2} \leq 3 - x$$
$$\Rightarrow \frac{x^2-9}{5x-2} + x - 3 \leq 0$$
$$\Rightarrow \frac{(x-3)(x+3)}{5x-2} + \frac{(x-3)(5x-2)}{5x-2} \leq 0$$
$$\Rightarrow \frac{(x-3)(x+3) + (x-3)(5x-2)}{5x-2} \leq 0$$
$$\Rightarrow \frac{((x+3) + (5x-2))(x-3)}{5x-2} \leq 0$$
$$\Rightarrow \frac{(6x+1)(x-3)}{5x-2} \leq 0$$

$\therefore p = 6$ and $q = 1$

17 b)
Critical values at $x = -\frac{1}{6}$, $x = \frac{2}{5}$ and $x = 3$

$x < -\frac{1}{6}$	$-\frac{1}{6} < x < \frac{2}{5}$	$\frac{2}{5} < x < 3$	$x > 3$
$-$	$+$	$-$	$+$

$$x \in (\infty, -\frac{1}{6}] \cup (\frac{2}{5}, 3]$$

18 a)
$\frac{(x-2)(x-6)}{x-8} \geq 0$ so $a = 2, b = 6, c = 8$

18 b)
$\frac{(x+3)(x-5)}{x-2} \geq 0$ so $a = -3, b = 5, c = 2$

19 a)
$\frac{(4-x)(x+2)}{x+4} \geq 0$ so $a = 4, b = -2, c = -4$

19 b)
$\frac{(7-x)(x-5)}{x-6} \geq 0$ so $a = 7, b = 5, c = 6$

20 a)

$$\frac{(x+1)(x+2)}{(x+3)(x+4)} < 0$$

Critical values at $x = -4$, $x = -3$, $x = -2$ and $x = -1$

$x < -4$	$-4 < x < -3$	$-3 < x < -2$	$-2 < x < -1$	$x > -1$
$+$	$-$	$+$	$-$	$+$

$$x \in (-4, -3) \cup (-2, -1)$$

20 b)

$$\frac{(x+2)(4-x)}{(x-1)(x+3)} \geq 0$$

Critical values at $x = -3$, $x = -2$, $x = 1$ and $x = 4$

$x < -3$	$-3 < x < -2$	$-2 < x < 1$	$1 < x < 4$	$x > 4$
$-$	$+$	$-$	$+$	$-$

$$x \in (-3, -2] \cup (1, 4]$$

20 c)

$$\frac{(x-3)(5-x)}{(2-x)(6-x)} \leq 0$$

Critical values at $x = 2$, $x = 3$, $x = 5$ and $x = 6$

$x < 2$	$2 < x < 3$	$3 < x < 5$	$5 < x < 6$	$x > 6$
$-$	$+$	$-$	$+$	$-$

$$x \in (-\infty, 2) \cup [3, 5] \cup (6, \infty)$$

21)

$e^x + 2 > 2$ for all real values of x

$$\frac{x+5}{2x-2} > 2$$
$$\Rightarrow \frac{x+5}{2(x-1)} > 2$$
$$\Rightarrow \frac{x+5}{x-1} > 4$$
$$\Rightarrow \frac{x+5}{x-1} \times (x-1)^2 > 4 \times (x-1)^2$$
$$\Rightarrow (x+5)(x-1) > 4(x-1)^2$$
$$\Rightarrow (x+5)(x-1) - 4(x-1)^2 > 0$$
$$\Rightarrow ((x+5) - 4(x-1))(x-1) > 0$$
$$\Rightarrow (9-3x)(x-1) > 0$$
$$\Rightarrow (3-x)(x-1) > 0$$

Critical values at $x = 1$ and $x = 3$

$x < 1$	$1 < x < 3$	$x > 3$
$-$	$+$	$-$

$$\{x : x \in \mathbb{R}, 1 < x < 3\}$$

22 a)

$$\frac{2x}{x-k} > x + k$$
$$\Rightarrow \frac{2x}{x-k} \times (x-k)^2 > (x+k) \times (x-k)^2$$
$$\Rightarrow 2x(x-k) > (x+k)(x-k)^2$$
$$\Rightarrow 2x(x-k) - (x+k)(x-k)^2 > 0$$
$$\Rightarrow (2x - (x+k)(x-k))(x-k) > 0$$
$$\Rightarrow (-x^2 + 2x + k^2)(x-k) > 0$$

Critical values at $x = 1 - \sqrt{k^2+1}$, $x = k$ and $x = 1 + \sqrt{k^2+1}$

$x < 1 - \sqrt{k^2+1}$	$1 - \sqrt{k^2+1} < x < k$	$k < x < 1 + \sqrt{k^2+1}$	$x > 1 + \sqrt{k^2+1}$
$+$	$-$	$+$	$-$

$$\{x : x \in \mathbb{R}, x < 1 - \sqrt{k^2+1}\} \cup \{x : x \in \mathbb{R}, k < x < 1 + \sqrt{k^2+1}\}$$

22 b)

$$\frac{x-4}{k-x} < 4-x$$
$$\Rightarrow \frac{x-4}{k-x} \times (k-x)^2 < (4-x) \times (k-x)^2$$
$$\Rightarrow (x-4)(k-x) < (4-x)(k-x)^2$$
$$\Rightarrow (x-4)(k-x) - (4-x)(k-x)^2 < 0$$
$$\Rightarrow (x-4)(k-x) + (x-4)(k-x)^2 < 0$$
$$\Rightarrow (1+(k-x))(x-4)(k-x) < 0$$
$$\Rightarrow (k+1-x)(x-4)(k-x) < 0$$

Critical values at $x = 4$, $x = k$ and $x = k+1$

$x < 4$	$4 < x < k$	$k < x < k+1$	$x > k+1$
$-$	$+$	$-$	$+$

$$\{x : x \in \mathbb{R}, x < 4\} \cup \{x : x \in \mathbb{R}, k < x < k+1\}$$

22 c)

$$\frac{x^2-k^2}{x+3} \geq k-x$$
$$\Rightarrow \frac{x^2-k^2}{x+3} \times (x+3)^2 \geq (k-x) \times (x+3)^2$$
$$\Rightarrow (x^2-k^2)(x+3) \geq (k-x)(x+3)^2$$
$$\Rightarrow (x^2-k^2)(x+3) - (k-x)(x+3)^2 \geq 0$$
$$\Rightarrow (x-k)(x+k)(x+3) + (x-k)(x+3)^2 \geq 0$$
$$\Rightarrow ((x+k)+(x+3))(x-k)(x+3) \geq 0$$
$$\Rightarrow (2x+k+3)(x-k)(x+3) \geq 0$$

Critical values at $x = -\frac{k+3}{2}$, $x = -3$ and $x = k$

$x < -\frac{k+3}{2}$	$-\frac{k+3}{2} < x < -3$	$-3 < x < k$	$x > k$
$-$	$+$	$-$	$+$

$$\{x : x \in \mathbb{R}, -\frac{k+3}{2} \leq x < -3\} \cup \{x : x \in \mathbb{R}, x \geq k\}$$